

Lecture 23

STAT 109: Introductory Biostatistics

Lecture 23: Organizing our Statistics Toolbox

Learning objectives

- Compute a test statistic using the normal approximation to the binomial and use it to perform a test (often called a **z-test**); in R, `prop.test()` in one line also gives a confidence interval.
 - Use organizational tools: what type of data you have and the purpose of the analysis.
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Variables and tools (overview)

- **Variables** — types of variables
 - **Descriptive statistics** — by number and type of variable
 - **Inferential statistics** for one binary variable: confidence interval vs. testing
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Standardizing statistics used to conduct a hypothesis test

Last lecture we saw the **z-score** as a way to compare relative standing in two different groups or within a group, assuming the reference group is normally distributed.

Let's apply the same idea of shifting and scaling to conducting a hypothesis test of a population proportion or probability p .

If $np \geq 10$ and $n(1 - p) \geq 10$, then the normal distribution can be used to approximate the **binomial** distribution of X = number of successes in n trials, where p is the probability of success in a single trial and the trials are independent.

On Midterm 1, you used this fact along with the empirical rule that 95% of the values of a normal distribution are within 2 SDs of the mean. We computed the mean of X as $\mu = np$ and the SD of X as $\sigma = \sqrt{np(1 - p)}$, and then rejected the null hypothesis if our observed value x was below $\mu - 2\sigma$ or above $\mu + 2\sigma$.

Another way to conduct this test is to **standardize** the observed value x by subtracting the mean and dividing by its standard deviation (also called **standard error** in this context):

Definition: standardized test statistic

$$\frac{\text{observed statistic} - \text{mean of sampling distribution}}{\text{standard deviation of sampling distribution}}$$

In the case of the statistic being a count of successes X from a binomial random process with n and p , the standardized test statistic is

$$z = \frac{x - np}{\sqrt{np(1 - p)}}.$$

This value can be compared with the standard normal distribution; a null hypothesis is rejected if z is below -2 or above 2 (the standard normal distribution has mean 0 and SD 1).

Equivalently, a standardized test statistic can be computed using the observed proportion $\hat{p} = x/n$: it is the same as the one based on x , but all terms are divided by n :

$$z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$$

Note that p is the value of the population proportion or probability hypothesized under the null hypothesis, and $\hat{p}$ is the observed proportion from n trials.

Example: Test whether a coin is fair if 16 heads were observed in 20 flips using a significance level of 0.05. Use a standardized test statistic and assume it follows the standard normal distribution.

In one line of R:

```
prop.test(x = 16, n = 20, p = 0.5)
```

```
1-sample proportions test with continuity correction
```

```
data: 16 out of 20, null probability 0.5
X-squared = 6.05, df = 1, p-value = 0.01391
alternative hypothesis: true p is not equal to 0.5
95 percent confidence interval:
 0.5573138 0.9338938
sample estimates:
 p
0.8
```

Interpretation: By default the test is **two-sided** (the alternative is that the coin is not fair). The output includes not only the p-value but also the lower and upper bounds of a confidence interval.

Organizing our statistics toolbox

We’ve now seen quite a few tools for analyzing data. Let’s think about how to keep them organized. It’s important to consider what sort of data we **have** and the purpose of the analysis — what we **want**.

Examples of what we might want

The purpose of a **descriptive statistics** analysis is to describe relevant information contained in a particular dataset.

If our data are binary outcomes (e.g. successes/failures, yes/no, good/bad), we have described them using:

- Counts
- Proportions

Example: Five students were asked: what is the maximum number of hot dogs you could eat?

Review notation: x , n , and $\hat{p} = x/n$ for count, sample size, and observed proportion.

If our data are a set of numbers, we summarize the **center** with the mean and the **spread** with a standard deviation.

In general, the type of calculation we use to describe a dataset depends on the *type* of information we have. Here are two useful definitions:

Definitions

A **variable** is a characteristic that may vary from object to object or person to person in our population of interest.

We’ll consider two **types** of variables:

- **Categorical** (qualitative) — distance or order between values is not meaningful.
- **Numerical** (quantitative) — distance and order are meaningful.

Examples: Classify each of the following variables as categorical or numerical:

- Age of a student in years
- Year a student was born
- Number of hot dogs eaten at a picnic
- Favorite beach of a student
- Rating on a scale of 1–5 for how much a student enjoys a statistics class

Steps in a descriptive analysis

1. Think about the purpose of your description. Decide what is “relevant.”
2. Classify each variable as either categorical or numerical.
3. If the variable is numerical, summarize its center and spread using mean and SD (more options will be discussed in the next lecture).
4. If the variable is categorical, summarize it using counts and proportions.

Inferential tools

If we want to infer the value of a parameter in the population from sample data, we currently have two **inferential** tools: confidence intervals and tests.

Confidence intervals answer: “What is a range of values that are reasonable for the parameter?” **Tests** compare the parameter against a **hypothesized** value.

We already have tests and confidence intervals for a population proportion/probability p ; these apply when the data are binary categorical.

We’ll spend the next few weeks expanding these tools to cover the cases of one numerical variable, two categorical variables, one numerical and one categorical variable, and two numerical variables.

We’ll also see a **third** inferential tool: **prediction**.